

Rohan Thapliyal

Full Stack Developer & AI Infrastructure

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Education

- 2026–2027 **Master of Science in Artificial Intelligence**, Oregon State University, Corvallis, OR
Accelerated Master's Program
- 2022–2026 **Bachelor of Science in Computer Science**, Oregon State University, Corvallis, OR, GPA: 3.92
Coursework: Software Engineering, Database Management Systems, Machine Learning, Artificial Intelligence

Experience

- Apr 2024–Apr 2026 **Full Stack Web Developer**, Tekbots, Corvallis, OR
- Architect, optimize, and maintain high-traffic university web tools and core platform services supporting thousands of active student and staff users.
 - Maintain robust SQL databases and backend microservices using PHP, Python, C++, and Bash to ensure reliable uptime and low-latency workflows.
- Sept 2024–Dec 2024 **Project Assistant Researcher**, OSU College of Agricultural Sciences, Corvallis, OR
- Conducted pre-grant research for the Agriculture Node Project, prototyping hardware and scripting systems for plant watering tracking devices.

Key Projects

- Present **Temporal Iterative Personal Project Assistant (TIPPA)**, AI Researcher & Developer
- Implement an actor-observer local AI agent (Python, Ollama) that develops "personality".
 - Utilize **Retrieval-Augmented Generation (RAG)** and **Vector Databases (ChromaDB)** for age-based memory compression and semantic retrieval.
 - Author research documentation and AI safety policy suggestions addressing risks of AI misuse, detection of model hallucinations, and mitigation strategies for psychological dependencies (AI psychosis).
- Present **Runes**, Mobile Full Stack Developer
- Co-develop a social media and astrology application utilizing React Native, TypeScript, Expo, and MongoDB.
 - Architect a non-obtrusive recommendation algorithm that models user interest patterns purely through direct post interactions, reducing invasive tracking common in traditional social platforms.
 - Build scalable **RESTful APIs** to generate personalized user feeds and natal chart graphics based on vectorized demographic interactions.
- Past **OpenBeavs**, DevOps Lead
- Design and manage continuous deployment pipelines using Cloud Build and Artifact Registry.
 - Containerize FastAPI backends and architect multi-agent AI protocols using Google Cloud Run within a **Microservice Architecture**.
- Past **OSU Course Data Collection Tool (DCT)**, Lead Full Stack Developer
- Developed and maintained the mandatory institutional assessment platform utilized university-wide by Oregon State University faculty to evaluate course outcomes and ABET accreditation metrics.
 - Built modular PHP backend services, high-throughput file upload pipelines for student submissions, and optimized relational SQL schemas to automate rubric evaluations and reviewer assignments.

Technical Skills

- Languages Python, C/C++, PHP, SQL, TypeScript, JavaScript, Java
- Frameworks React, Node.js, FastAPI, Flask, Expo
- Cloud & DevOps Google Cloud Platform (Cloud Run, Vertex AI), Docker, CI/CD
- AI & Data PyTorch, TensorFlow, Ollama, ChromaDB, PostgreSQL, MongoDB
- Hardware Arduino, ESP32, Embedded Systems